Experiment - 7

Flow control at Data Link Layer using Sliding window protocol

Aim:

To write a program to implement flow control at the Data link Layer using the Sliding window protocol.

Algorithm:

1) Start the program
2) Read the total number of frames (N), window size (W), the frame number to be lost.
3) Initialize:
   base = 0 - first window ledged frame
   next seq = 0 - next frame to be sent
4) While base < N, determine the sender window from base to base + W - 1
5) Send all frames within the current window and increment next seq after sending each frame.

6) If the specified frame is lost
   * Receiver deletes the missing frame
   * Receiver discards the out-of-order frame
   * sender experience a timeout

7) If there is no frame loss:
   * Receiver accepts the frames in order
   * Receiver sends an Ack for each received frame.

8) Repeat steps 4-7 until all frames are Successfully transmitted acknowledged

9) Display that all frames

10) stop the program

Code:-

```
import java.util.Scanner;
public class Sliding window {
static void simulate (int total frames,
int window size, Integer lost frame){
    int base = 0;
    int nextSeq = 0;
while (base < total frames){
    Window End = Math.min(
    Window Size
```

```
System.out.println ("sender window : ( frame" + base
                    "-frame" + windowEnd + ")");
while (nextSeq < base + windowSize && nextSeq <
                                    totalFrames) {
    System.out.println ("Node A -> sending frame"
                        + nextSeq);
    if (lostFrame != null && nextSeq == lostFrame)
    {
        System.out.println ("Frame" + nextSeq +
                            "Lost in transit!");
        nextSeq;
    }
}
if (lostFrame != null && base <= lostFrame
    && lostFrame < nextSeq) {
    System.out.println ("Node B -> Frame" +
                        lostFrame + "missing");
    System.out.println (
        "Node A -> Time out ! Resending
        window starting from frame" +
                        lostFrame + ":");
    nextSeq = lostFrame;
    lostFrame = = null;
    continue;
    for (int j = base; j < nextSeq;
                        j++) {
```

```
System.out.println(
        "Node B -> Received frame " + ft + ", Sending
         Ack " + f );
    }
    base = next seq;
  }
}
public static void main (String[] args) {
    Scanner sc = new Scanner (System.in);
System.out.println (
    "Enter total number of frames to send ");
int totalFrames = sc.nextInt();
System.out.println (
    "Enter frame number to simulate
                as Lost \n" +
    "(0 to " + (totalFrames - 1) +
        enter -1 for no lost
    );
}else {
    lostFrame = input lost frame;
}
Simulate (totalFrame, windowSize,
                lostFrame);
    sc.close ();
}
}
```

output

Enter total number of frames to send : 3

Enter Senders window size : 2

Enter frame number to simulate as Lost
(0 to 2, enter -1 for no lost) : 1

Sender window : [Frame 0 - Frame 1]

Node A → Sending Frame 0

Node B → Sending frame 1

Node B → Frame 1 Lost in transmit)

Node A → Timeout ! Resending frame
frame 1

Sender window : [Frame 1 - frame 2]

Node A → Sending frame 1

Node A → Sending frame 2

Node B → Recived frame 1, Sending Ack 1

All 3 frames sucessfully transmitted from
node A to Node B

Result:

Thus the program to implement flow control at
the data link layer using the sliding window
protocol was implemented sucessfully.